# Methodology

RealFact: OpenMP-Accelerated Real-Time Multi-Source News Verification

IT 301 — Parallel Computing


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This methodology explains how the RealFact system will be built and tested to answer the project's core question: can large-scale, multi-source evidence retrieval for real-time news verification be executed efficiently on multicore CPUs using OpenMP, without duplicate or syndicated reports being miscounted as independent evidence? It follows the pipeline: what data will be used → how it will be collected → how it will be processed → what method will be applied → how it will be evaluated.


## I. Dataset Explanation

*A. Datasets Used.* Two categories of data are used: (i) three established fact-verification benchmarks, used to measure verification quality against published baselines, and (ii) a live retrieval corpus, used to demonstrate and benchmark real-time operation. Table I summarizes both.

The three benchmarks play distinct and non-redundant roles. FEVER [2] contributes the largest labeled set and the standard retrieve-then-classify formulation the project builds on — what [1] formalizes as the evidence-retrieval and verdict-prediction stages of automated fact-checking. This project addresses only those two stages: claim detection and justification production, the framework's other components, are out of scope. LIAR [3] checks that verdicts generalize to short, informally worded political statements rather than only Wikipedia-style claims. FakeNewsNet [4] is included because, unlike FEVER and LIAR, it consists of full real-world news articles with source URLs and publication metadata — which lets duplicate and syndication detection actually be exercised, since FEVER and LIAR contain no syndicated copies of the same story. *FakeNewsNet is therefore the only dataset on which the independence filter can be evaluated at all*, and this asymmetry is treated as a designed contrast rather than an inconvenience (Section III-D). The live retrieval corpus is what the OpenMP retrieval engine searches against at query time; it is unlabeled, since its only role is to supply candidate evidence and to provide a corpus large enough for the scaling study.

*B. Label and Task Harmonization.* The three benchmarks do not natively share a label space, and one of them does not natively express a claim–evidence relation at all. The mapping is stated explicitly so that reported accuracy numbers are interpretable:

- **FEVER** is used as-is: SUPPORTED → SUPPORT, REFUTED → CONTRADICT, NOT ENOUGH INFO → NEUTRAL.
- **LIAR**'s six-way scale is collapsed to three: {true, mostly-true} → SUPPORT; {false, pants-fire} → CONTRADICT; {half-true, barely-true} → NEUTRAL. This last mapping is imperfect and is flagged as a known limitation. As [1] argues, mid-scale labels such as "half-true" are typically meta-ratings over composite claims whose parts differ in veracity, rather than a single intermediate truth value; such a claim is therefore closer to the system's *Conflicting Evidence* verdict (parts disagree) than to NEUTRAL (no evidence exists). Results on LIAR's middle band are reported separately from the two-sided bands.
- **FakeNewsNet** labels *articles*, not claim–evidence pairs. The article headline is taken as the claim, the article body is excluded from the corpus for that query, and the label is mapped real → SUPPORT, fake → CONTRADICT. Because "fake" means "unsupported by the record" rather than "actively contradicted," CONTRADICT and NEUTRAL are additionally scored as a merged *not-supported* class, and both figures are reported.

*C. Data Preprocessing.*

- **Text cleaning:** HTML/boilerplate stripping, whitespace and encoding normalization, and language filtering (English-only for this phase), applied uniformly to benchmark articles and live-feed articles so that no preprocessing difference can masquerade as a domain effect.
- **Deduplication keys:** each article is hashed on (source domain, normalized headline) as a cheap first-pass exact-duplicate filter, ahead of the embedding-based near-duplicate clustering of Section II-C, to avoid wastefully embedding exact re-posts.
- **Embedding generation:** claims and articles are encoded with BGE-small-en-v1.5 [5] into $d = 384$ dimensional unit-normalized vectors; article embeddings are pre-computed and cached at ingestion, so query time is spent on comparison, not encoding. BGE-small truncates at 512 tokens, so a long article is represented by its lead — which aids syndication detection (syndicated copies share leads) but can miss evidence buried deep in a body. This is recorded as a known ceiling on Recall@$K$, not silently absorbed.
- **Leakage control:** for every query drawn from a labeled benchmark, the source article itself and any article sharing its deduplication key are removed from the retrieval corpus before search. Without this, the system retrieves the article under test and trivially "supports" its own headline, and reported accuracy becomes meaningless.
- **Splitting:** FEVER and LIAR use their official train/dev/test splits; FakeNewsNet, which has no official split, is partitioned 70/10/20 by article, stratified by real/fake label *and* grouped by syndication cluster, so that copies of one story cannot straddle the train and test sides.

TABLE I
Datasets used for verification quality and real-time retrieval

| Dataset | Role | Size | Labels / Notes |
|---|---|---|---|
| FEVER | Verification quality (train/tune) | 185,445 claims (145,449 train / 19,998 dev / 19,998 test); evidence drawn from ∼5.4M Wikipedia documents (∼25M candidate sentences) | SUPPORTED, REFUTED, NOT ENOUGH INFO |
| LIAR | Verification quality (generalization check) | 12,836 short political statements; 80/10/10 split (∼10,269 / 1,284 / 1,283) | 6-way: pants-fire, false, barely-true, half-true, mostly-true, true (collapsed to 3-way) |
| FakeNewsNet | Verification quality (real-world news; *only* source of syndication) | 23,196 news items (PolitiFact 432 fake / 624 real + GossipCop 5,323 / 16,817, 2014–2018) with source URLs, following the count tabulated in [1]; link rot reduces the number actually retrievable at crawl time | Binary real / fake, inherited from PolitiFact and GossipCop editorial verdicts; the size actually retrieved is reported alongside results |
| Live Retrieval Corpus | Real-time evidence pool (unlabeled) | Continuously updated; target of tens of thousands of articles resident at any time, extensible to 500K for the scaling study | Title, body text, publish timestamp, source domain — pulled from permitted RSS feeds / news APIs |

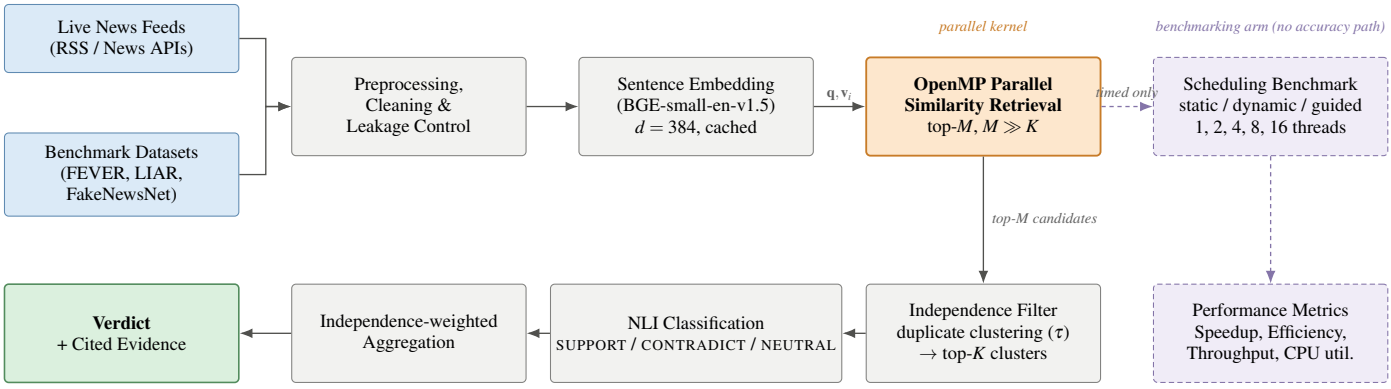



Fig. 1. End-to-end system pipeline. The verification pipeline runs left to right along the top row and back along the bottom row; the dashed benchmarking arm branches off the retrieval stage and measures parallel performance only, so the systems claim and the accuracy claim are falsifiable independently.

## II. Proposed Method

*A. System Overview.* Fig. 1 gives the end-to-end view as a single top-to-bottom pipeline. The two input sources feed a shared preprocessing and embedding stage; the OpenMP-parallel retrieval and independence-filtering stages sit at the center of the verification pipeline; and a separate, clearly-branched benchmarking arm measures the parallel performance of the retrieval stage itself, independent of the verification pipeline's accuracy. The two arms share code but are reported separately, because a systems contribution and an accuracy contribution have to be falsifiable independently.

*B. Retrieval and Candidate Generation.* Let the corpus be $\{a_i\}_{i=1}^{N}$ with cached unit-normalized embeddings $\mathbf{v}_i \in \mathbb{R}^d$, $d = 384$, and let a claim $c$ embed to $\mathbf{q}$. Because all vectors are unit-normalized, cosine similarity reduces to a dot product:

$$s_i = \cos(\mathbf{q}, \mathbf{v}_i) = \mathbf{q}^\top \mathbf{v}_i. \tag{1}$$

The scan over $i$ is exhaustive and is the parallel kernel of the project. A brute-force scan — rather than an approximate index — is a deliberate choice: the object of study is the parallel scaling of an exact, deterministic kernel, and an ANN index would change *what* is retrieved between configurations, confounding the speedup measurement with a recall change. An ANN comparison is noted as future work, not as a baseline.

*C. Duplicate Clustering and the Retrieve-Wide Rule.* Outlets frequently republish the same underlying report, so the raw ranked list is dominated by near-identical copies. Critically, clustering is applied to a *wide* candidate pool, not to the final $K$: the parallel scan returns the top $M$ candidates with $M \gg K$ (default $M = 10K$), these $M$ are clustered by pairwise embedding similarity above a threshold $\tau$, and only then are the top $K$ *clusters* selected and their representatives passed downstream. Clustering after truncation to $K$ would be self-defeating — if ten outlets syndicate one story and $K = 10$, the top-$K$ contains ten

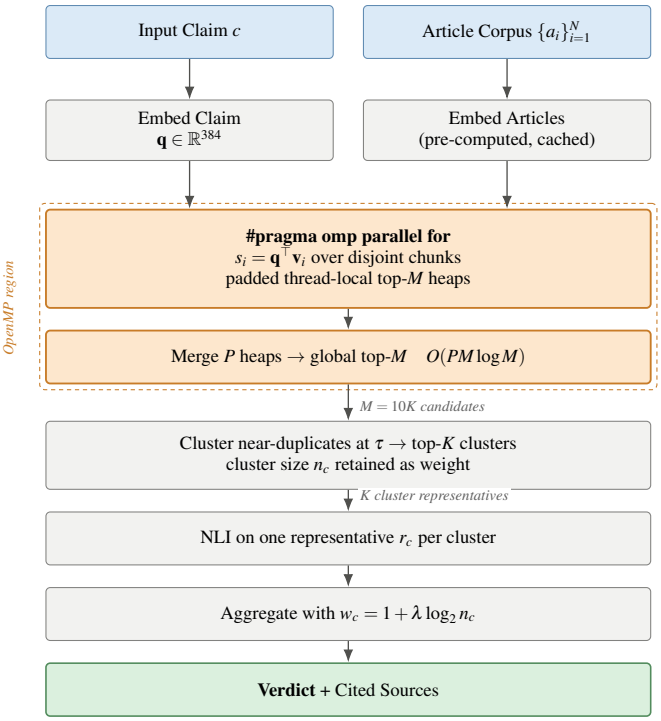



Fig. 2. Proposed method. The order of operations is the load-bearing detail: retrieve *wide* ($M \gg K$), then cluster, then keep the top-$K$ *clusters* — not top-$K$ articles followed by clustering.

copies of a single report, and collapsing them leaves nine empty slots and no additional evidence. Retrieving wide is what makes the independence filter load-bearing rather than cosmetic. Fig. 2 shows the resulting order of operations.

*D. Independence-Weighted Aggregation.* The NLI model labels each cluster representative $r_c$ with respect to the claim, yielding a signed stance $\sigma_c \in \{+1, -1, 0\}$ for SUPPORT/CONTRADICT/NEUTRAL and a confidence $p_c \in [0,1]$. A cluster of $n_c$ syndicated copies is not $n_c$ independent witnesses; neither is it exactly one, since independent outlets sometimes converge on the same wording. The cluster weight therefore interpolates between the two extremes:

$$w_c \;=\; 1 + \lambda \log_2 n_c, \qquad \lambda \in [0,1], \tag{2}$$

so that $\lambda = 0$ gives strict one-vote-per-cluster and $\lambda \to 1$ approaches naive duplicate-counting. The aggregate evidence score and the supporting-side fraction are

$$S \;=\; \frac{\sum_c w_c\, p_c\, \sigma_c}{\sum_c w_c\, p_c}, \qquad D \;=\; \frac{\sum_{c:\sigma_c \neq 0} w_c\, p_c\, [\sigma_c = +1]}{\sum_{c:\sigma_c \neq 0} w_c\, p_c}, \tag{3}$$

where $S$ is the net signed stance and $D$ is the fraction of non-neutral weight on the supporting side. The four-way verdict follows from thresholds $\theta$ (evidence sufficiency) and $\delta$ (disagreement):

| Condition | Verdict |
|---|---|
| $\sum_{c:\sigma_c \neq 0} w_c p_c < \theta$ | Unverified |
| $\delta < D < 1 - \delta$ | Conflicting Evidence |
| $S \geq 0$, otherwise | Supported |
| $S < 0$, otherwise | Contradicted |

Retaining *Conflicting Evidence* as a first-class verdict rather than forcing a binary call is deliberate. Most fact-checking pipelines assume a single authoritative source, and handling mutually contradicting or untrustworthy evidence is named in [1] as an open research challenge; reporting disagreement is a more honest output than a coin-flip verdict on a genuinely contested claim.

For scoring against the three-way benchmarks, Supported → SUPPORT, Contradicted → CONTRADICT, and both Unverified and Conflicting Evidence → NEUTRAL; the two are also reported disaggregated, since collapsing them hides the behaviour the independence filter is meant to produce.

$\lambda$, $\tau$, $\theta$ and $\delta$ are tuned on training/dev splits only. Note the honest constraint: FEVER and LIAR contain no syndication, so $\tau$ and $\lambda$ cannot be tuned on them. They are set on a small manually-annotated subset of FakeNewsNet training articles in which syndication clusters are labeled by hand, and the sensitivity of final verdicts to both is reported as a sweep rather than asserted as a tuned optimum.

*E. Parallelization Strategy.* The scan is parallelized with `#pragma omp parallel for` [6] over the corpus index. Each thread maintains a private top-$M$ min-heap, padded to a cache-line boundary to prevent false sharing on the heap metadata; thread-local heaps are merged into the global top-$M$ in a single serial reduction whose cost is $O(PM \log M)$ and therefore independent of $N$. Embeddings are stored as one contiguous row-major $N \times d$ `float` array so that each thread's chunk is a unit-stride stream.

The scheduling clause (static / dynamic / guided) and the thread count are treated as configurable parameters rather than fixed choices, since comparing them is one of the project's stated contributions. *The cost of comparing one article is a fixed $d$-element dot product and does not depend on the original article's length* — length variance is eliminated at embedding time — so any observed scheduling difference must be attributed to chunk-boundary cache behaviour, heap-update frequency (chunks containing many high-scoring articles do more heap work), OS jitter, and dynamic scheduling's atomic loop-counter overhead, not to input-size imbalance.

## III. Experimental Settings and Analysis

*A. Platform and Baselines.*

- **Platform disclosure:** physical core count, SMT status, socket and NUMA-node count, cache sizes, memory channels and peak STREAM bandwidth, compiler and flags, and OpenMP runtime version are all reported. Threads are pinned (`OMP_PROC_BIND=close`, `OMP_PLACES=cores`); without pinning, 16-thread numbers on an 8-core machine are uninterpretable. Turbo/boost state is stated explicitly, since

a single-threaded baseline runs at a higher clock than a fully-loaded one and silently deflates measured speedup.

- **Serial reference $T_1$:** a build compiled *without* OpenMP, so that runtime and reduction overheads are not charged to the baseline. The one-thread OpenMP build is timed as well and reported separately, and the gap between them is stated as the OpenMP entry cost.
- **Accuracy baselines:** (i) the serial version of the same retrieval-and-NLI pipeline, to confirm that parallelization changes *how* retrieval is computed and not *what* is retrieved — an exact bit-level match of retrieved IDs between serial and every parallel configuration is asserted as a correctness test, with tie-breaking made deterministic by index; (ii) published FEVER baselines [2] as an external accuracy reference.
- **Ablations:** the independence filter is run in three modes — off, one-vote-per-cluster ($\lambda = 0$), and log-weighted — so its contribution is isolated from the rest of the pipeline.

*B. Measurement Protocol.*

- **Scheduling $\times$ thread sweep:** static, dynamic and guided at 1, 2, 4, 8 and 16 threads (15 configurations) on the retrieval stage, with dynamic and guided additionally swept over chunk size.
- **Repetitions and reporting:** each configuration is run 11 times after 2 discarded warm-up runs; the median and interquartile range are reported. Latency distributions are right-skewed and bounded below, so the mean is not the right summary.
- **Significance:** a Wilcoxon signed-rank test on paired per-query latencies is used instead of a paired $t$-test, since the normality assumption does not hold for latency data. Effect size is reported alongside the $p$-value; with speedups in the $4\times$ range the practical significance is carried by the effect size, not the test.
- **Corpus-size scaling:** the retrieval stage is timed against corpora of 10K / 50K / 100K / 500K articles at fixed thread count, to separate scalability from the thread sweep. Working-set size ($N \times d \times 4$ bytes — 768 MB at $N = 500$K, $d = 384$, so far beyond any L3) is reported next to each point and marked against the L3 capacity, since the interesting inflection is the cache-to-DRAM crossing.
- **Real-time target:** "real-time" is given a number — end-to-end p95 latency per claim at a stated corpus size and thread count — rather than left as an adjective.

*C. Evaluation Metrics.* Verification quality and parallel performance are judged on two independent axes, matching the project's two claimed contributions.

*Verification quality.* Macro and per-class precision, recall and F1 on SUPPORT/CONTRADICT/NEUTRAL, plus overall accuracy, on FEVER, LIAR and FakeNewsNet. Evidence-retrieval Recall@$K$ is reported on FEVER (which has annotated evidence sentences) and on FakeNewsNet (where the article's cited source URLs are treated as gold), but *not* on LIAR, which has no annotated evidence documents and for which the metric is undefined.

*Parallel performance.* Speedup $S_p = T_1/T_p$, parallel efficiency $E_p = S_p/p$, throughput in queries per second, and CPU utilization, across scheduling strategies and corpus sizes. Arithmetic intensity is reported alongside: the kernel performs 2 flops per 4-byte element loaded, i.e. 0.5 flop/byte, which places it firmly in the memory-bound region of the roofline [7] and predicts the ceiling that the speedup curve should hit.

The approach is considered successful if it matches or approaches published FEVER-style accuracy while showing a statistically significant, near-linear speedup up to the bandwidth-imposed ceiling — not perfect linear speedup.

*D. Expected Analysis.*

- **Speedup** should rise close to linearly at low thread counts (1–4) and then flatten. The flattening is expected to begin *before* the physical core count is exhausted, because at 0.5 flop/byte the kernel saturates memory bandwidth before it saturates cores. Attributing the knee to core count alone would be the wrong explanation; the predicted knee is where measured bandwidth meets the STREAM ceiling. SMT threads beyond the physical core count are expected to add little, since there is no arithmetic latency to hide.
- **Scheduling.** Because per-article cost is constant, static is expected to win on lower overhead, with dynamic and guided losing ground through atomic counter contention at small chunk sizes. Any advantage for dynamic or guided would indicate an *external* imbalance source — co-scheduled OS work, NUMA-remote pages, or frequency variation across cores — and would be investigated as such rather than reported as load imbalance from the data.
- **Independence filter.** Expected to measurably change verdicts on FakeNewsNet, which contains real syndicated duplicates, and to have little or no effect on FEVER and LIAR, which do not. That contrast is itself the validation: a filter that changed FEVER verdicts would be removing genuine evidence, not duplicates. The retrieve-wide rule is expected to be a precondition for any effect at all; with clustering applied after top-$K$ truncation, the effect should vanish, and this is run as an ablation to demonstrate it.
- **Verification accuracy.** Expected to be comparable to, not better than, existing FEVER-style pipelines, since the contribution is systems-level efficiency and duplicate-awareness rather than a novel classification model. The expected win is latency and scalability at equal or near-equal accuracy. Accuracy is expected to be lowest on LIAR, whose short claims carry little retrievable context.
- **Scalability.** As corpus size grows, per-thread throughput is expected to degrade gradually rather than catastrophically, with a visible step where the embedding matrix stops fitting in L3. The shape of that degradation curve — not a single-point speedup number — is what characterizes scalability.

*E. Threats to Validity.* Four are acknowledged in advance. First, headline-as-claim on FakeNewsNet is a proxy for the annotated claims the dataset does not provide, and it biases the task toward headline-level checkability. Second, syndication clusters are identified by embedding similarity, which cannot distinguish

syndication from independent convergence on a wire-service fact; the reported effect of the independence filter is therefore an upper bound on true redundancy correction. Third, models trained on FEVER are known to key on indicative claim words rather than on the retrieved evidence [1], so a strong FEVER number alone would not establish that retrieval quality is doing the work — part of why FakeNewsNet and LIAR are reported beside it. Fourth, the live corpus changes between runs, so all timing results are taken against a frozen snapshot, and the snapshot's size and date are reported.

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